Muscle Coactivation Levels Reflect a Tradeoff Between Metabolic Cost and Performance When Responding to Physical Disturbances During Upper Limb Posture Control

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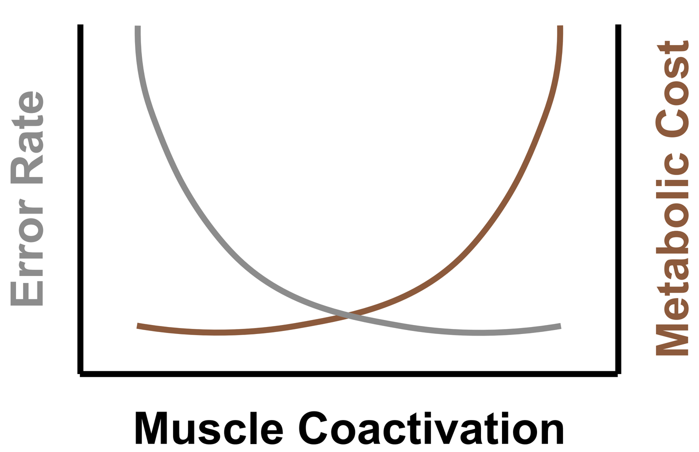
Introduction

Muscle coactivation - the simultaneous activity of agonist and antagonist muscles - has been shown to improve motor performance by enabling muscles to generate forces more rapidly and respond more effectively to sensory feedback. However, this enhanced performance comes at a cost, and the metabolic expense associated with muscle coactivation remains unknown in the context of upper limb posture control. The main objective of this study is to investigate the trade-off between metabolic cost and performance under different levels of muscle coactivation.

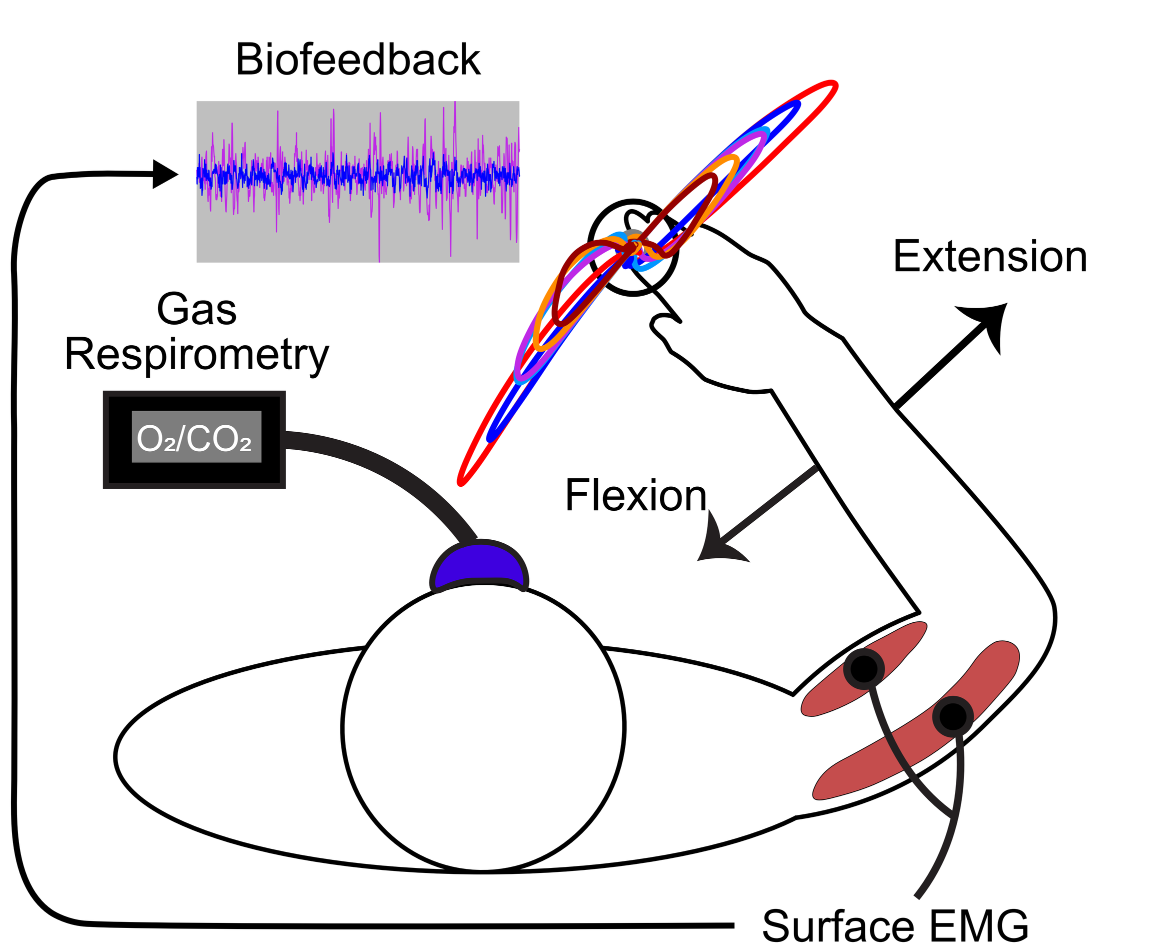
Hypothesis & Prediction

We hypothesize that there is a trade-off between muscle coactivation, metabolic cost, and task performance. We predict that higher levels of coactivation will lead to improved performance at the expense of increased metabolic cost.



Methods

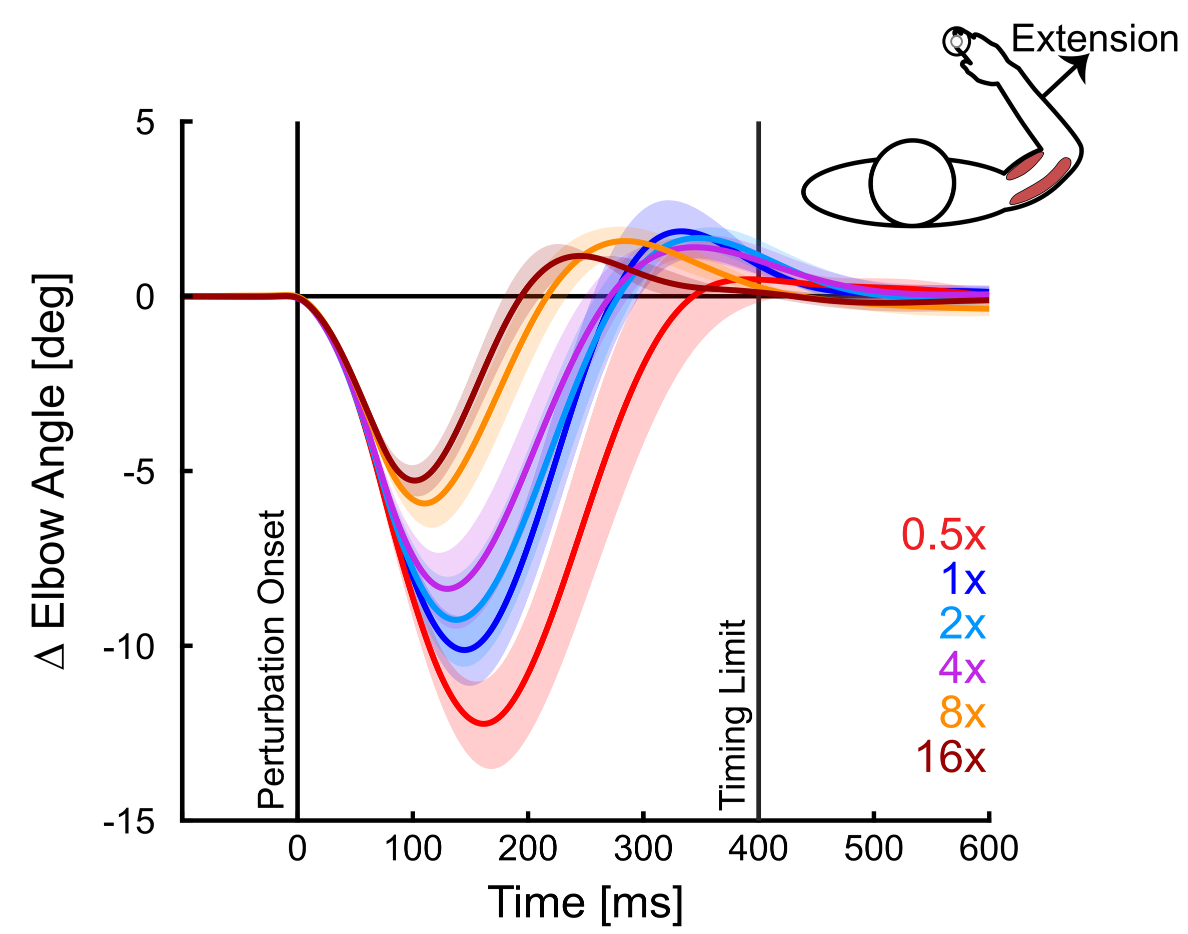
Five healthy participants attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances.



Participants performed the task under self-selected coactivation levels. Biofeedback was used to maintain coactivation at [0.5, 2, 4, 8, 16]x their self-selected level, with testing continued until failure.

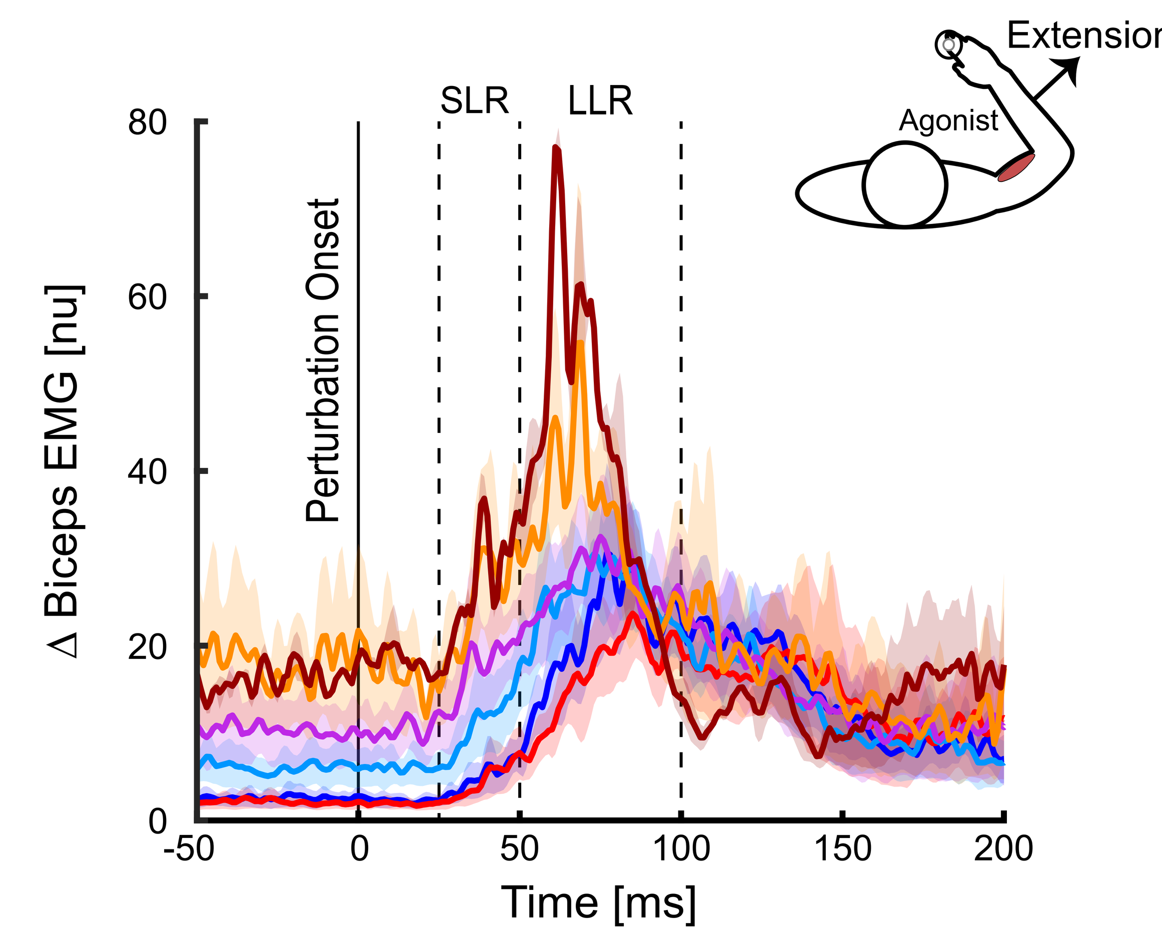
Results

Coactivation Reduces Peak Displacement

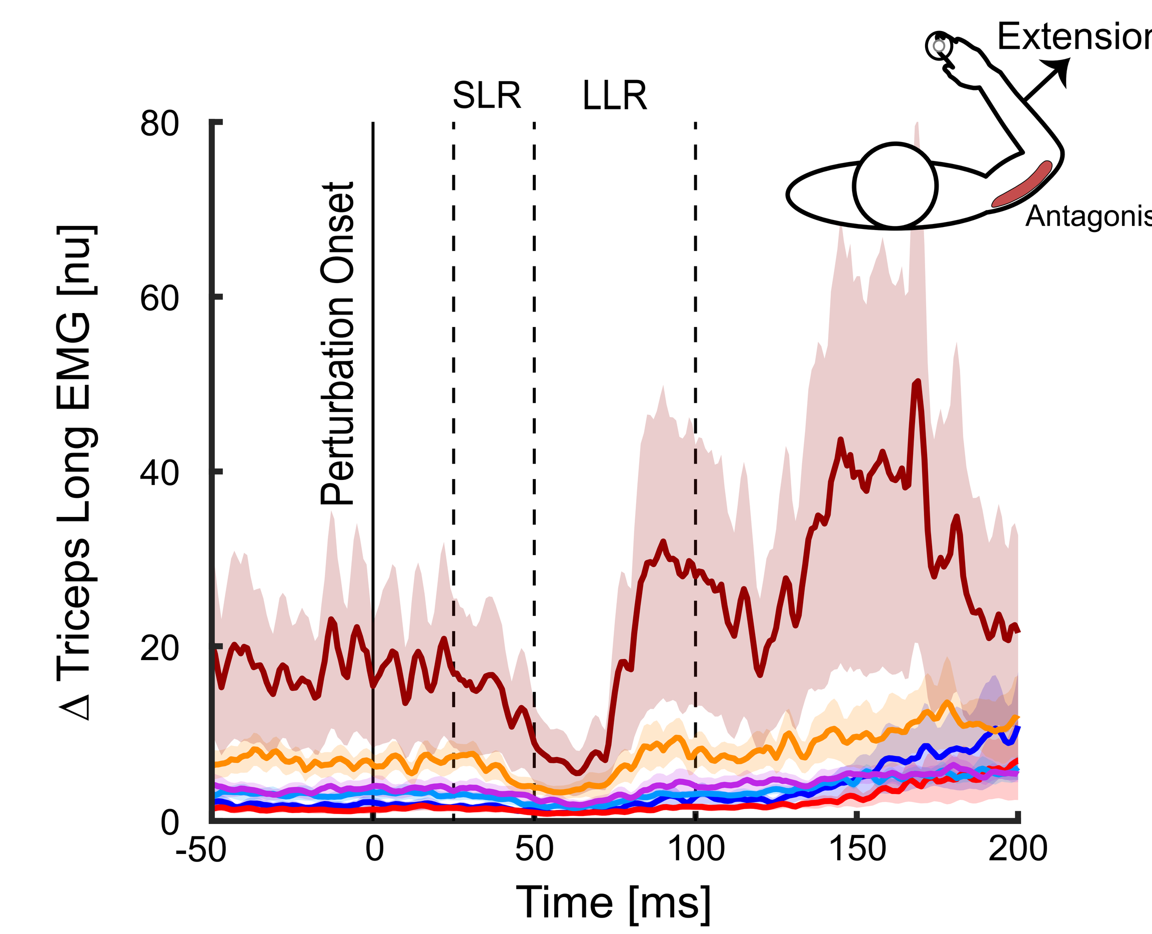


Peak displacements scale with the level of muscle coactivation.

Muscle Coactivation Increases Responsiveness to Proprioceptive Feedback

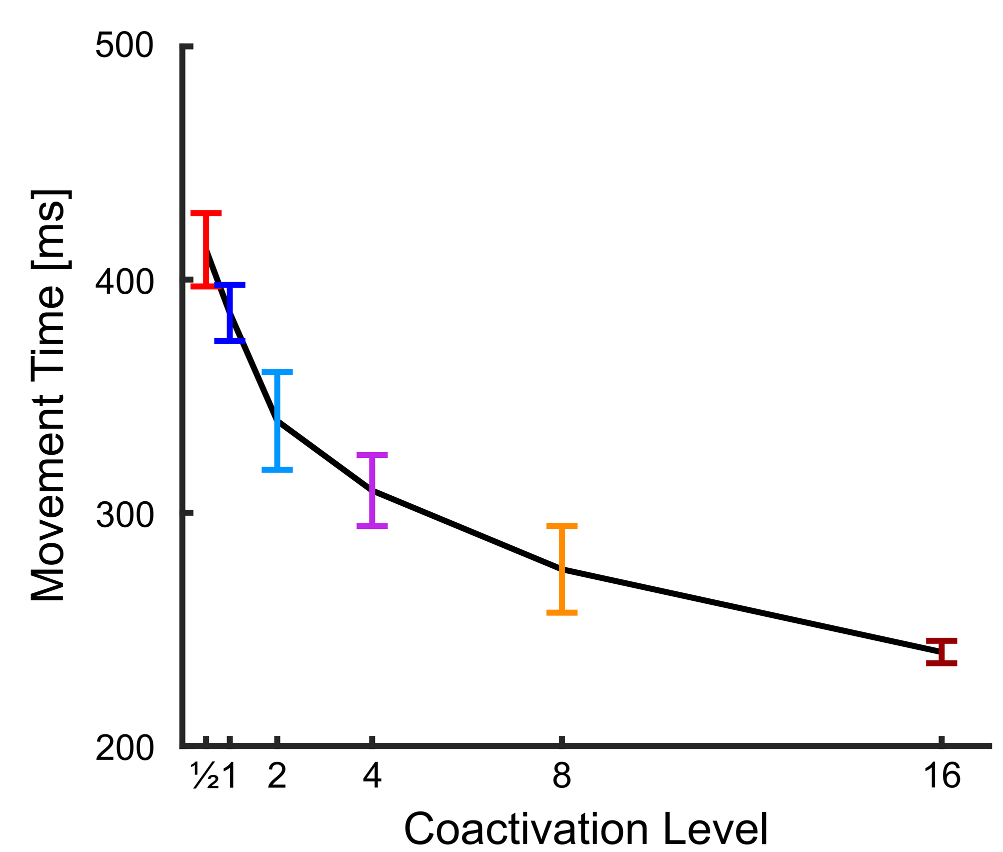


Increased muscle coactivation results in greater stretch responses of agonist muscle.



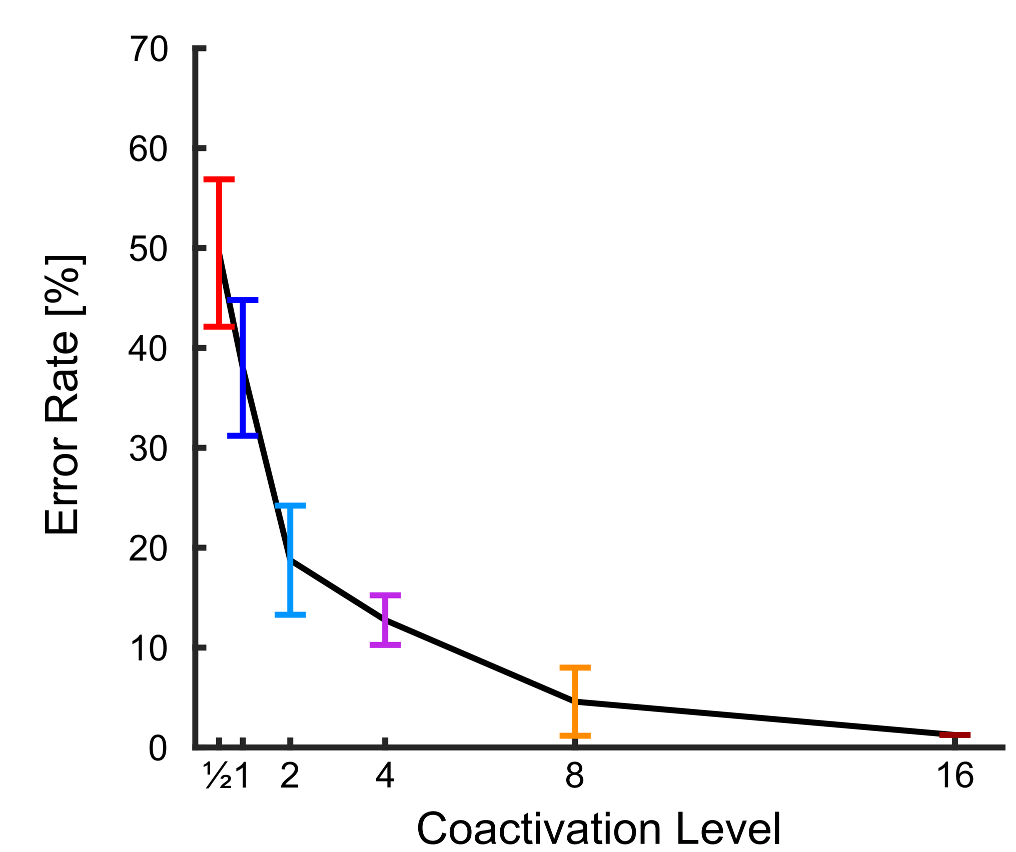
Increased muscle coactivation enables more inhibition of antagonist muscle.

Coactivation Facilitates Faster Movements

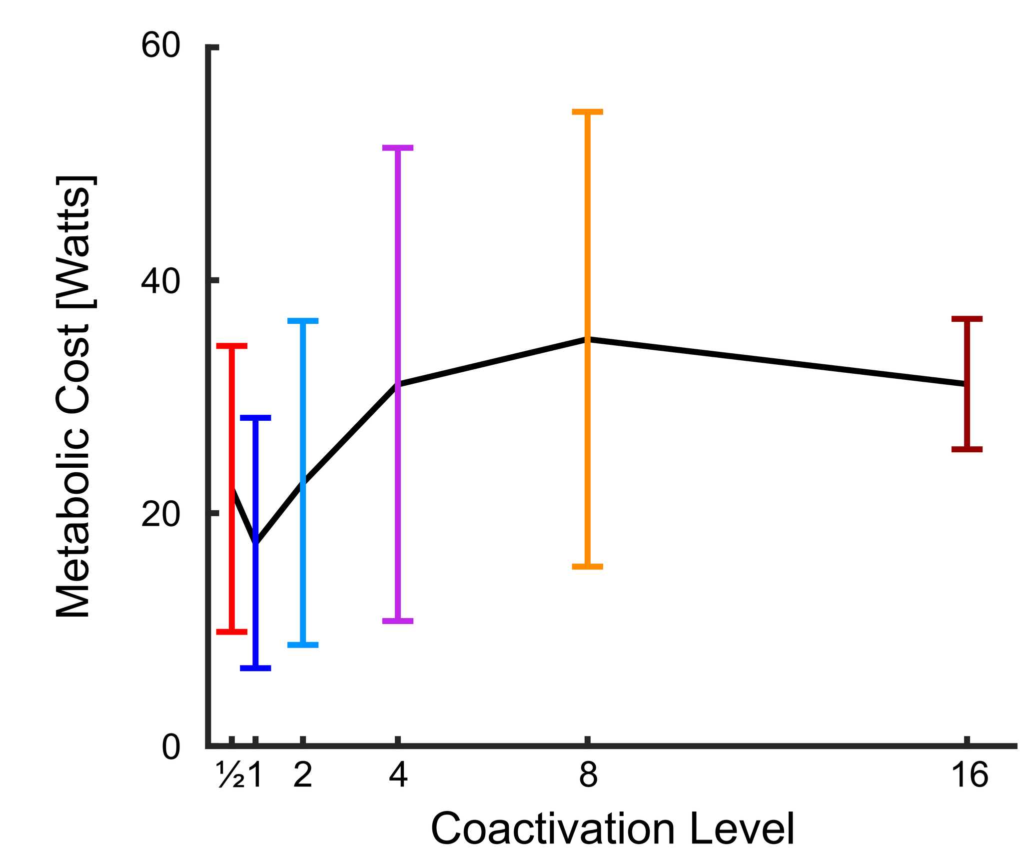


Higher muscle coactivation reduces movement time after perturbation.

High Coactivation Improves Performance But Increases Metabolic Cost



Higher muscle coactivation reduces error rates in disturbance responses.



Higher muscle coactivation levels lead to increased metabolic cost.

Discussion

At spontaneous levels of muscle coactivation, the nervous system prioritizes the minimization of metabolic cost at the expense of performance. However, increasing the level of coactivation via biofeedback-controlled muscle activity improves performance at the expense of metabolic cost. This research offers insights into the role of muscle coactivation in responding to sensory feedback.

Conclusion

There is a complex interplay between metabolic cost and performance in the control of human arm movements and postures.